ISM Note 6

Yang Chen

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1. Energy Balance and Temperature

A photon with $\nu > \nu_1$ can ionize the HI, and the excess energy will be converted into kinetic energy of free electron, and then the electrons will share the energy with ions. This is the physical picture of photoionization heating.

The heating rate per HI is given by

$$\Gamma = \int_{\nu_1}^{\infty} \frac{4\pi J(\nu)}{h\nu} \sigma_{\rm ion}(\nu)(h\nu - h\nu_1) d\nu \sim \varphi \langle h(\nu - \nu_1) \rangle$$

and then heating rate per volume is

$$H \sim n_{\rm HI}\, \varphi \langle h(\nu - \nu_1) \rangle$$

We assume ionization balance, so we have

$$H \sim n_{\rm HI}\, \varphi = n^2 x^2 \alpha(T)$$

The recombination cooling rate is similarly given by

$$\mathcal{L}_{\rm rec} = n_{\rm HI} \sum_i \int \sigma_{\rm rec,i}(v)\, v f(v) (\frac{1}{2} m_e v^2) dv \sim n^2 x^2 \alpha(T) k_B T$$

We consider the energy balance here, so the cooling must balance the heating, we then have

$$H \sim \mathcal{L}_{\rm rec}$$

This reduces to

$$k_B T \sim \langle h(\nu - \nu_1) \rangle$$

We plug in the numbers and we can get $T \sim 1.6 \times 10^5 \mathrm{K} > 10^4 \mathrm{K}$. This temperature is much higher than the real temperature, which means whether the heating is overestimated or the cooling is underestimated.

For this reason, we must include heavy element cooling, and the balance relation becomes

$$H(T) = \mathcal{L}_{\mathrm{rec}}(T) + \sum_x \mathcal{L}_x(T)$$

Here the heavy element cooling is denoted by $\mathcal{L}_x(T)$.

Note that since here we assume $H \propto n^2$ and $\mathcal{L} \propto n^2$, the equilibrium temperature of HII region is independent of density!

Multiphase ISM Equilibrium

ISM cools via radiation, and this requires the ISM to be optically thin.

Not to confuse the reader, we need to clarify here that **the equilibrium does not mean stable**, they are different concepts. Though we assume the thermal equilibrium, it can still be thermally unstable. And now we are going to discuss the thermal stability of the ISM. This is important since the thermal instability is directly related to the existence of ISM gas phases.

The net energy loss rate is given by

$$\mathcal{L}_{\mathrm{net}} = n^2 \Lambda(T) - n^2 \Gamma(T)$$

and from this we can derive the stability criterion, which is if

$$\frac{\partial \mathcal{L}}{\partial T} > 0$$

then it is stable, otherwise it is unstable.

This can be easily understood through a simple analysis for the response of the ISM to perturbation. If the stability criterion is satisfied, then if we have a positive temperature perturbation, the net energy loss will increase, this will cause the gas to cool down back to its original temperature, so it is stable. The case for negative temperature perturbation is similar.

A sketch for the $\mathcal{L}(T)$ is shown as follows,

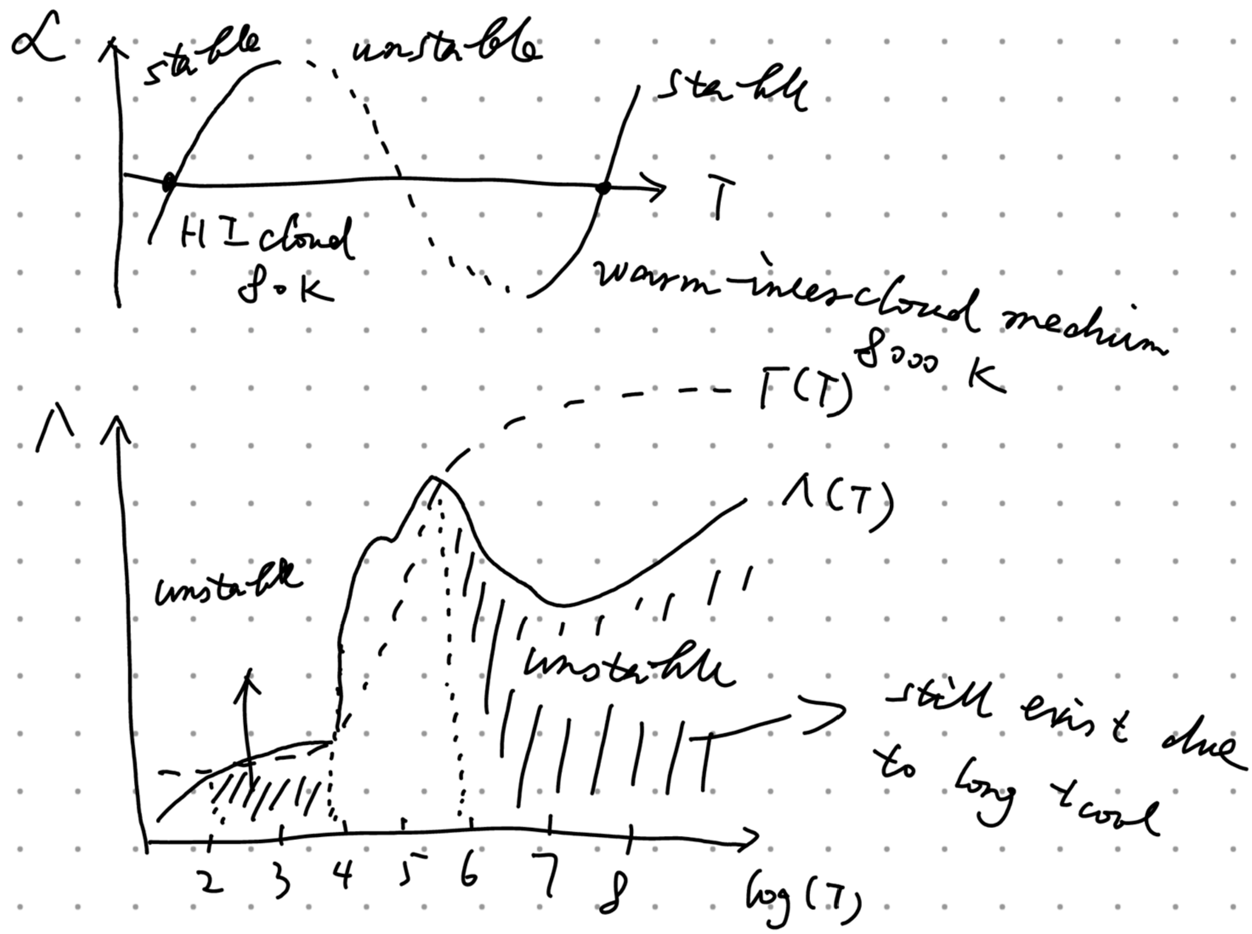

Fig 1. Stability diagram for ISM.

From this figure, we can see that the intermediate temperature range $10^4\mathrm{K} < T < 10^6\mathrm{K}$ is stable, while others are unstable. So the multiphase in the title literally means the coexistence of gases from $10^4\mathrm{K}$ to $10^6\mathrm{K}$.

But in the previous class we have mentioned that the ICM gas has very high temperature, so according to our analysis above it should be thermally unstable. However, we can still observe them, so why for that?

One probable reason is, the cooling time is very long, recall that the cooling time is given by

$$t_{\mathrm{cool}} \sim \frac{k_B T}{n\Lambda(T)}$$

If the temperature is very high, and the density is very low, the cooling time can be very long. Therefore, although the gas is thermally unstable, it takes a long time for it to cool down, so the gas can still exist for a long time.

Another reason is, there may exist some feedback mechanism which can enhance the heating, but this is still in research and does not have a clear answer now. This problem is the so called "cooling flow problem".

Note that we assume the ISM is in thermal equilibrium, but this is **the global thermal equilibrium**, which means the heating and cooling are balanced globally, and they may not balance **locally**. This implies that locally the cooling rate and heating rate can be different, so our discussion above can make sense.

2. Hydrodynamics

The main topic here is **how can we describe the motion of a group of particles**. The tool we use is the phase-space density or distribution function. The distribution function is defined as

$$f(\vec{x}, \vec{p}) = \frac{dN}{d^3\vec{x}\, d^3\vec{p}}$$

First we have the Boltzmann equation, which is given by

$$\frac{\partial f}{\partial t} + \vec{v} \cdot \nabla_x f + \vec{F} \cdot \nabla_p f = 0$$

This equation tells us the change of the distribution function is caused by the free streaming and the collision. Physically, it describes **the conservation of particle numbers**.

Starting with the Boltzmann equation, the first thing we can do is to integrate it over momentum space, and we can get

$$\frac{\partial n}{\partial t} + \frac{\langle \vec{p} \rangle}{m} \frac{\partial n}{\partial x} = 0$$

where n is the number density, and $\langle \vec{p} \rangle$ is the average momentum.

This is **the continuity equation** or mass equation. It can be easily seen that this continuity equation has two variables, namely n and $\langle \vec{p} \rangle$, which can not be solved with only one equation. So we need another equation to solve for these two variables.

Then the second thing we can do is to multiply the Boltzmann equation by a factor of $\frac{\vec{p}}{m}$, and then integrate over momentum space, and we can get

$$\frac{\partial}{\partial t}\frac{\langle \vec{p} \rangle}{m} + \frac{\partial(\vec{v} \otimes \vec{v} + \langle \vec{w} \otimes \vec{w} \rangle)}{\partial x} - n\vec{g} = 0$$

where $\vec{w}$ is the random velocity, and

$$\vec{v} \otimes \vec{v} = \begin{pmatrix} v_x^2 & v_x\, v_y & v_x\, v_z \\ v_x\, v_y & v_y^2 & v_y\, v_z \\ v_x\, v_z & v_y\, v_z & v_z^2 \end{pmatrix}$$

By definition, we have

$$P = \frac{1}{3}\,\rho\,\mathrm{Tr}(\langle \vec{w} \otimes \vec{w} \rangle) = \frac{1}{3}\,\rho\,(w_x^2 + w_y^2 + w_z^2) = \rho\, w_x^2 = \rho\, w_y^2 = \rho\, w_z^2$$

And we define the momentum tensor as

$$\tilde{T} = \rho\,(\vec{v} \otimes \vec{v})$$

We assume the velocity distribution is isotropic so the pressure is isotropic as well. Then we can rewrite the stress tensor as

$$\tilde{T} = \rho\,\langle \frac{\vec{p} \otimes \vec{p}}{m^2} \rangle = \rho\,(\vec{v} \otimes \vec{v}) + P \cdot I + \tilde{T}_{\text{vis}}$$

where $\tilde{T}_{\text{vis}} = \rho\,(\vec{w} \otimes \vec{w})$ is the viscous tensor.

If $\langle w_x \cdot w_y \rangle = \langle w_x \rangle \langle w_y \rangle = 0$, in other words the random motion in three directions are statistically independent, then we can find

$$\tilde{T}_{\text{vis}} = 0$$

Then with the viscous tensor disappeared, finally we can get **the momentum equation**, which is given by

$$\frac{\partial}{\partial t}(\rho\,\vec{v}) + \nabla\cdot\tilde{T} = \rho\,\vec{g}$$

Hydrostatic equilibrium

The hydrostatic equilibrium means $\vec{v} = 0$, so from the above equation we have

$$\nabla P = \rho\,\vec{g}$$

As an example, we consider the accretion disk around the black hole, the gravity field is given by

$$\vec{g} = -\frac{GM}{r^2}\phi\,\hat{z}$$

and we assume the radial component is balanced by centrifugal force, e.g $g \sim \Omega^2 z$, then we can solve for the hydrostatic equation and get

$$\rho(z) = \rho_0\,e^{-z^2/2H^2}$$

where the scale height is given by $H = \frac{c_s}{\Omega}$

The scale height is **the characteristic height** of the disk. Physically speaking, if the sound speed is very high, which means the temperature is high, the scale height is large and the disk is thick. On the contrary, if the sound speed is low, which means the temperature is low so the thermal pressure is weak, the centrifugal force will dominate, so the scale height would be small and the disk is thin.

Euler Equation

Let's go back to the momentum equation, and we neglect the viscous term, then we have

$$\frac{\partial}{\partial t}(\rho\,\vec{v}) + \nabla\cdot(\rho\vec{v}\otimes\vec{v} + P\cdot I) = \rho\,\vec{g}$$

We plug in the continuity equation and do some simple calculations, then we can derive

$$\frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \nabla)\vec{v} = -\frac{1}{\rho}\nabla P + \vec{g}$$

Then we can define a so-called **Lagrangian derivative**, which is given by

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + (\vec{v} \cdot \nabla)$$

Then we can rewrite the above equation as

$$\frac{D\vec{v}}{Dt} = -\frac{1}{\rho}\nabla P + \vec{g}$$

This is **the Euler equation**, which describes the motion of a fluid. The detailed discusstion of this equation will be given in the next class.